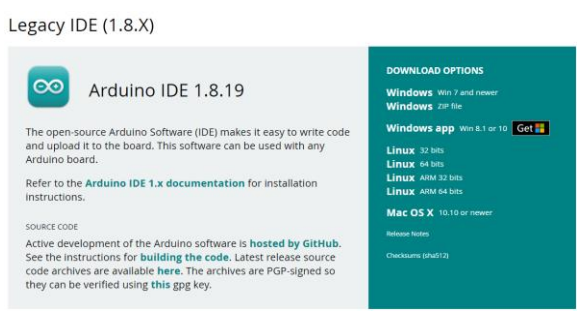
Getting Start ESP32: ESP32 GPIO + ESP32 Interface

1/10. Arduino IDE - 1.8.19 Install + Blink Test

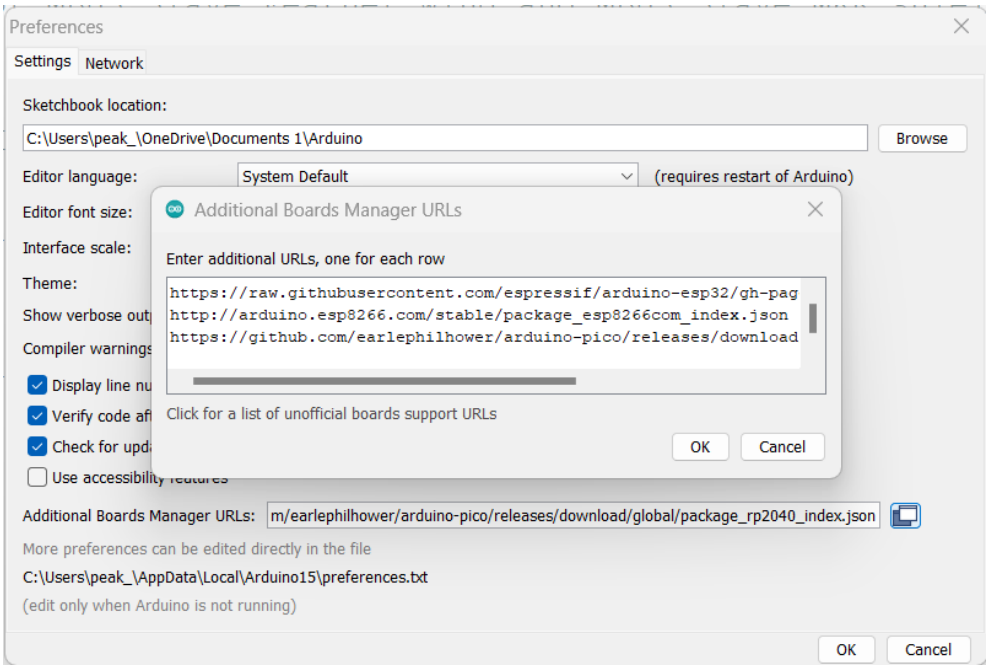
1. Read https://www.cybertice.com/p/5089
2. Read https://support.arduino.cc/hc/en-us/articles/360019833020-Download-and-install-Arduino-IDE
3. Download "arduino-1.8.19-windows.exe" and Install



Windows, Linux or Mac

Just Download

4. Add ESP32 Board URLs: File → Preference → <Addd 3 Line > → Ok → Ok



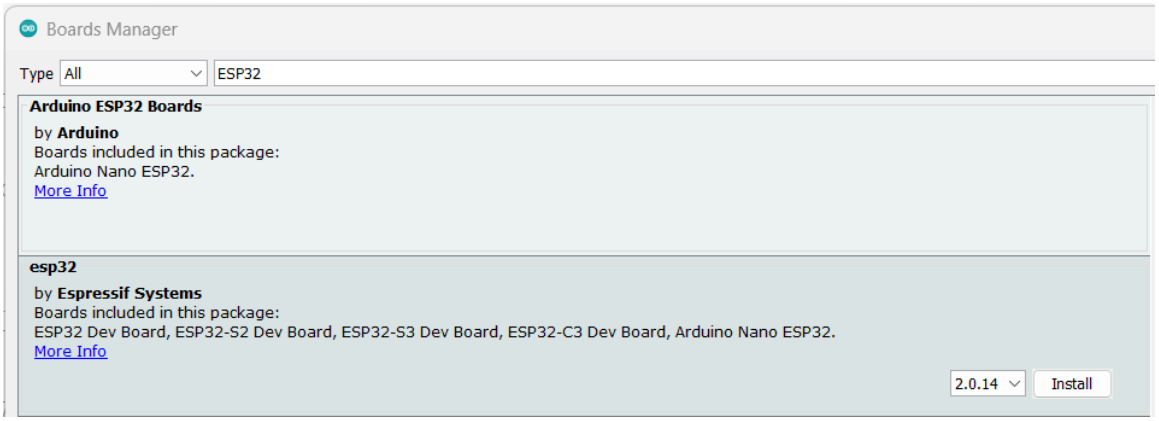
Additional Boards Manager URLs:

https://raw.githubusercontent.com/espressif/arduino-esp32/gh-pages/package_esp32_index.json

http://arduino.esp8266.com/stable/package_esp8266com_index.json

https://github.com/earlephilhower/arduino-pico/releases/download/global/package_rp2040_index.json

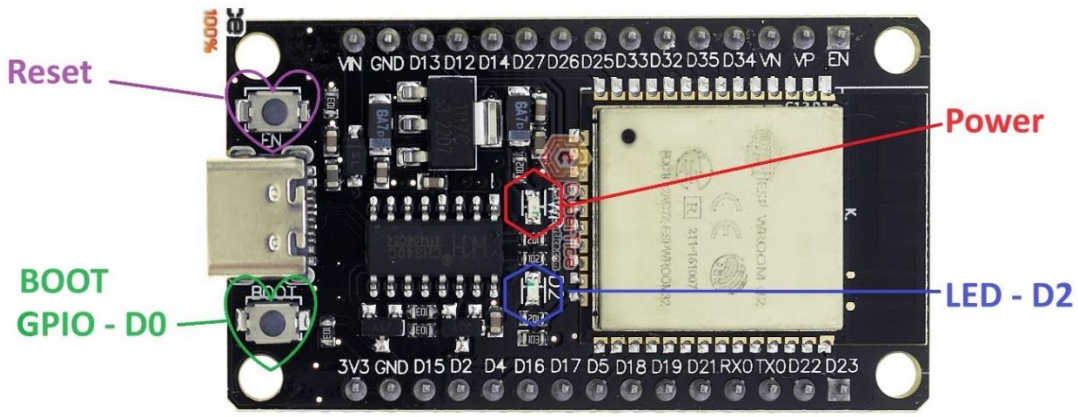
5. Add ESP32 Board: Tools → Boards → Board Manager



Filter = esp32

Select → esp32 by Espressif System → Version 2.0.14 → Install

6. Read https://www.allnewstep.com/b/144
7. Connect ESP32 Board to PC
8. Blink Test Code: File → Example → 01.Basic → Blink
9. Select Serial Port: Tools → Port → ComX
10. Select Target Board: Tools → Boards → ESP32 Arduino → DOIT ESP32 DEVKIT V1
11. Upload Code to Target Board: Sketch → Upload
12. ESP32 Pinout Reference https://randomnerdtutorials.com/esp32-pinout-reference-gpios/



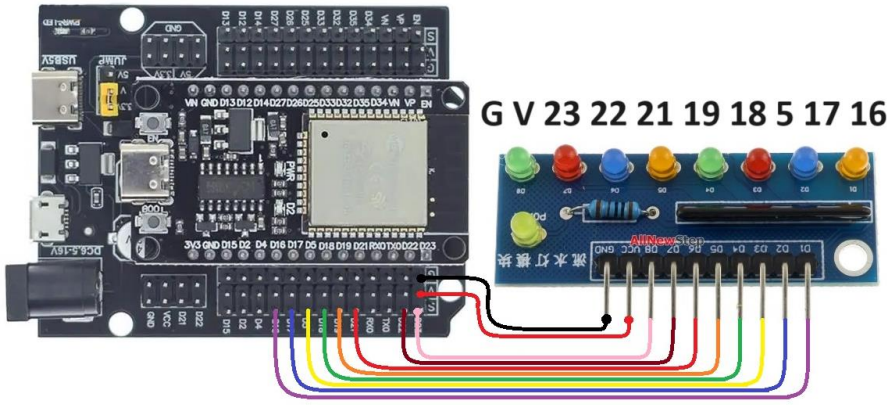
Blink Test: On Board LED = LED_BUILTIN = GPIO D2

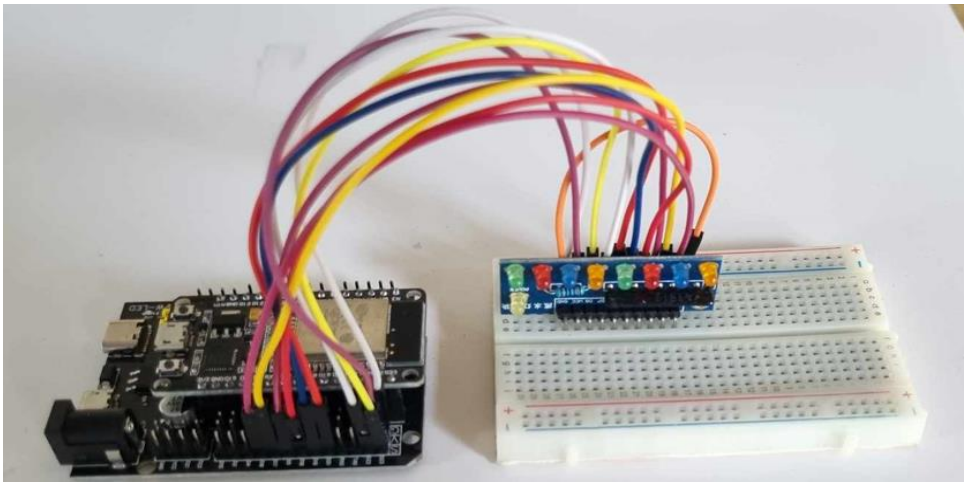
```
void setup() {
  pinMode(LED_BUILTIN, OUTPUT);
}

void loop() {
  digitalWrite(LED_BUILTIN, HIGH); delay(1000);
  digitalWrite(LED_BUILTIN, LOW);  delay(1000);
}
```

- ทดลองปรับค่าในฟังก์ชัน delay

2/10. ESP32 GPIO

13. Read https://www.allnewstep.com/b/157
14. Wirring “Circuit-02”





15. Test Code “Test0201-D23 Blink”

`void setup() {`
`  pinMode(23, OUTPUT);`
`}`
 	
`void loop() {`
`  digitalWrite(23, HIGH); delay(1000);`
`  digitalWrite(23, LOW);  delay(1000);`
`}`
	• fix 23

`#define Test_LED 23`
`void setup() {`
`  pinMode(Test_LED, OUTPUT);`
`}`
 	
`void loop() {`
`  digitalWrite(Test_LED, HIGH); delay(1000);`
`  digitalWrite(Test_LED, LOW);  delay(1000);`
`}`
	• #define

`int Test_LED = 23;`
`void setup() {`
`  pinMode(Test_LED, OUTPUT);`
`}`
 	
`void loop() {`
`  digitalWrite(Test_LED, HIGH); delay(1000);`
`  digitalWrite(Test_LED, LOW);  delay(1000);`
`}`
	• int variable

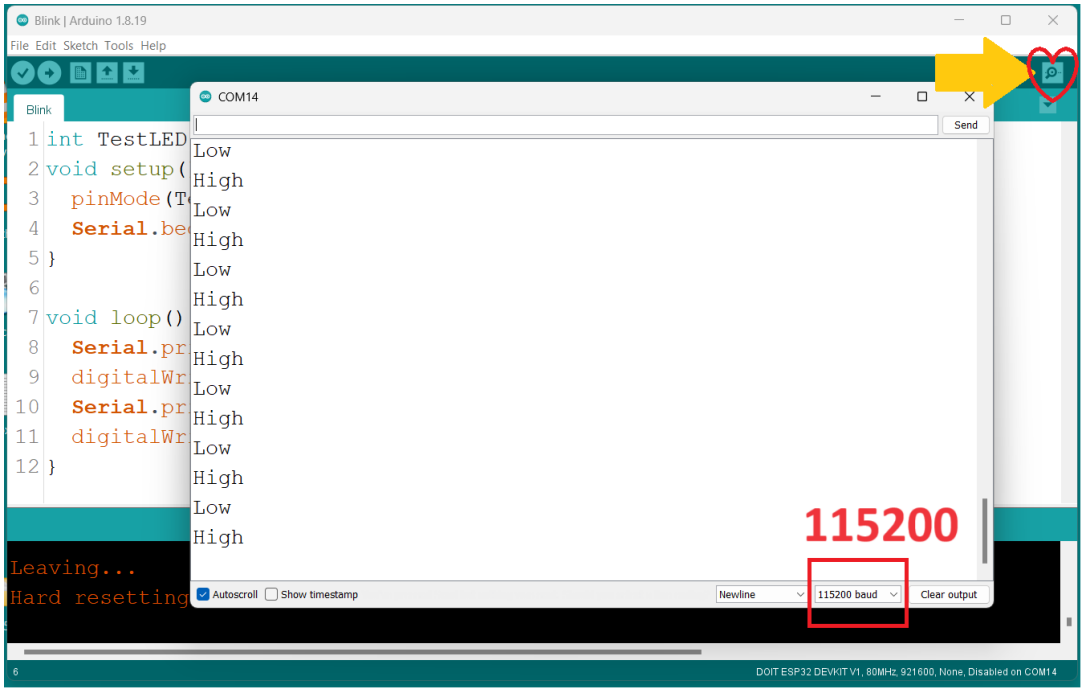
16. Test Code "Test0202-D23 Blink with Serial Monitor"

```
int TestLED = 23;
void setup() {
 pinMode(TestLED, OUTPUT);
 Serial.begin(115200);
}

void loop() {
 Serial.println("High");
 digitalWrite(TestLED, HIGH); delay(1000);
 Serial.println("Low");
 digitalWrite(TestLED, LOW);  delay(1000);
}
```

- Serial Baud rate = 115200

Tools → Serial Monitor

Set baud = 115200

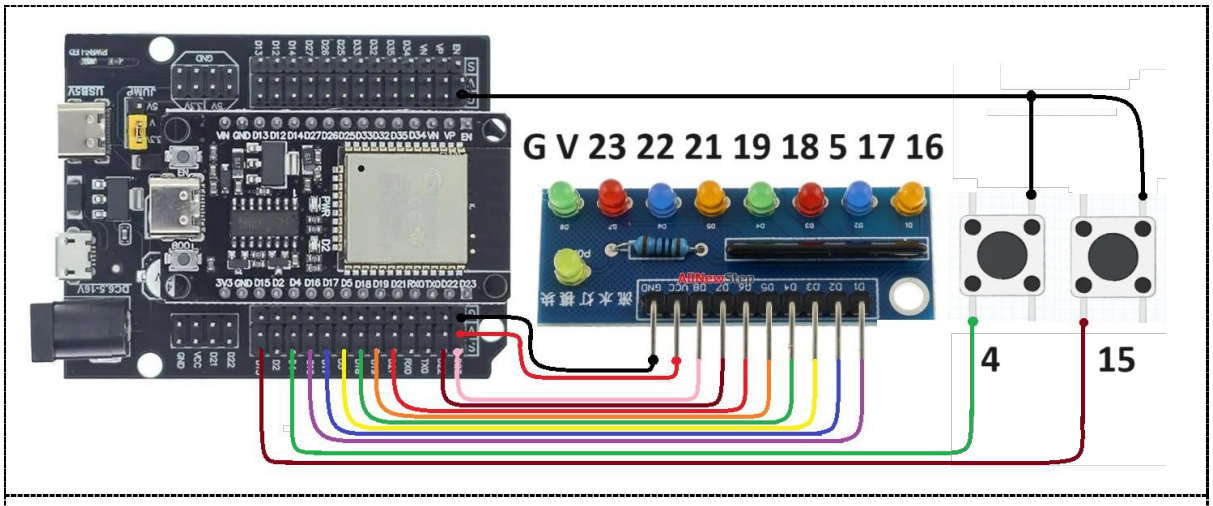


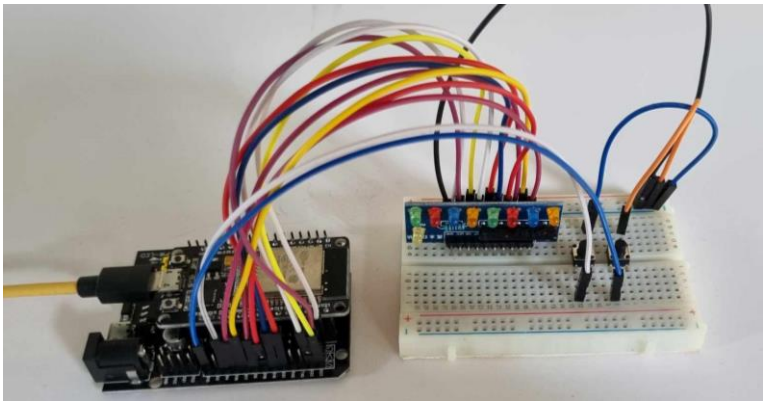
17. Test Code "Test0203-Moving LED"

```
int pinIO[] = {23, 22, 21, 19, 18, 5, 17, 16};

void setup() {
 for (int i = 0; i <= 7; i++) {
  pinMode(pinIO[i], OUTPUT);
  digitalWrite(pinIO[i], HIGH); // Active Low
 }
}

void loop() {
 for (int i = 0; i <= 7; i++) {
  digitalWrite(pinIO[i], LOW); // On LED
  delay(250);
  digitalWrite(pinIO[i], HIGH); // Off LED
 }
}
```

- Show L2R Moving
- How to show R2L Moving
- How to show R2L and L2R Moving

18. Add 2 Switch in “Circuit-02”

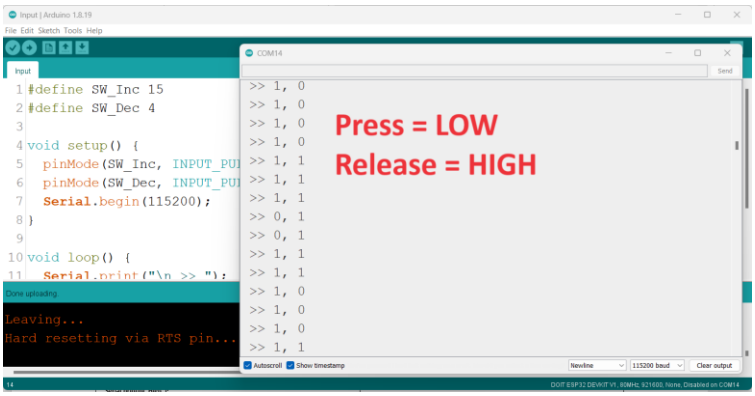


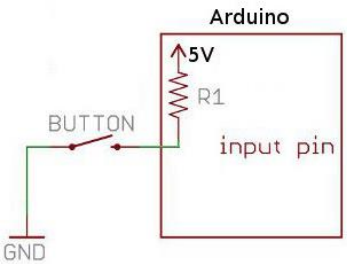


19. Test Code “Test0204-Input Switch”

```
#define SW_Inc 15
#define SW_Dec 4

void setup() {
  pinMode(SW_Inc, INPUT_PULLUP);
  pinMode(SW_Dec, INPUT_PULLUP);
  Serial.begin(115200);
}

void loop() {
  Serial.print("\n >> ");          Serial.print(digitalRead(SW_Inc));
  Serial.print(", ");               Serial.print(digitalRead(SW_Dec));
  delay(250);
}
```

- pinMode = INPUT, INPUT_PULLUP, OUTPUT

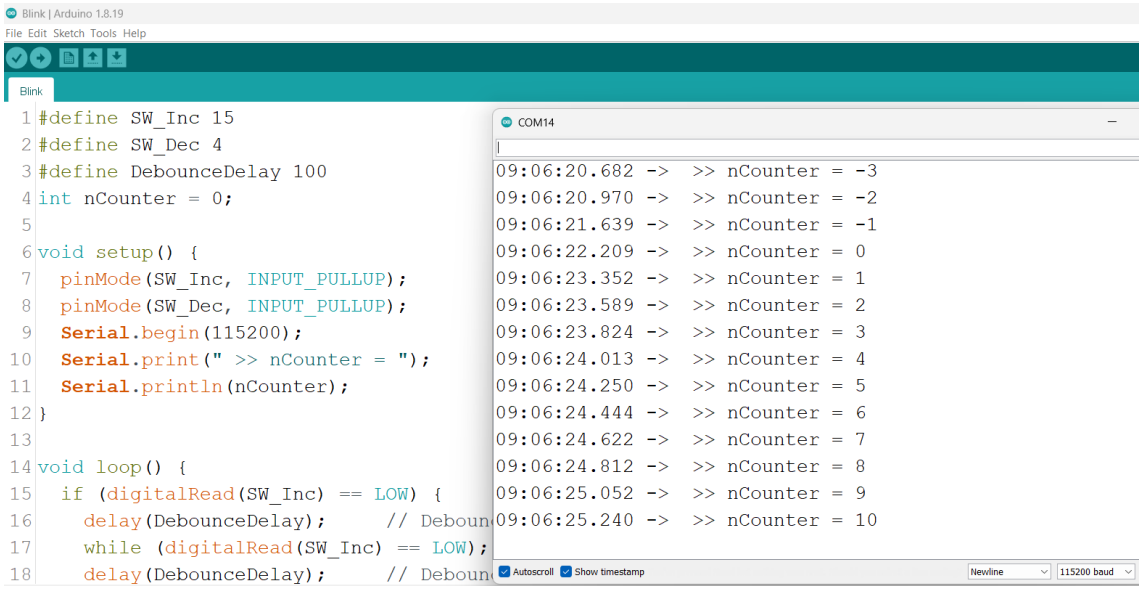




20. Test Code "Test0205-Push Counter"

```
#define SW_Inc 15
#define SW_Dec 4
#define DebounceDelay 100
int nCounter = 0;

void setup() {
  pinMode(SW_Inc, INPUT_PULLUP);
  pinMode(SW_Dec, INPUT_PULLUP);
  Serial.begin(115200);
  Serial.print(" >> nCounter = ");
  Serial.println(nCounter);
}

void loop() {
  if (digitalRead(SW_Inc) == LOW) {    // If Switch Press
    delay(DebounceDelay);    // Debounce Delay H->L
    while (digitalRead(SW_Inc) == LOW); // wail until release
    delay(DebounceDelay);    // Debounce Delay L->H
    nCounter++;
    Serial.print(" >> nCounter = ");
    Serial.println(nCounter);
  }

  if (digitalRead(SW_Dec) == LOW) {
    delay(DebounceDelay);
    while (digitalRead(SW_Dec) == LOW);
    delay(DebounceDelay);
    nCounter--;
    Serial.print(" >> nCounter = ");
    Serial.println(nCounter);
  }
}
```

- Test for changing pinMode
  pinMode(SW_Inc, INPUT);
- Test for short debounce delay
  #define DebounceDelay 1
- Test for long debounce delay
  #define DebounceDelay 10000



21. Test Code “Test0206-Moving LED with 2 Switch”

```
#define SW_L2R 15
#define SW_R2L 4
int pinIO[] = {23, 22, 21, 19, 18, 5, 17, 16};

void setup() {
  for (int i = 0; i <= 7; i++) {
    pinMode(pinIO[i], OUTPUT);
    digitalWrite(pinIO[i], HIGH); // Active Low
  }
  pinMode(SW_L2R, INPUT_PULLUP);
  pinMode(SW_R2L, INPUT_PULLUP);
  Serial.begin(115200);
}

int CountRun = 0, StepRun = 0;
int minIndex = 0, maxIndex = 7;
void loop() {
  if (digitalRead(SW_L2R) == LOW) StepRun = +1;
  if (digitalRead(SW_R2L) == LOW) StepRun = -1;
  CountRun += StepRun;  // Next Step
  CountRun = CountRun < minIndex ? maxIndex : CountRun;
  CountRun = CountRun > maxIndex ? minIndex : CountRun;
  Serial.print("\n > ");
  Serial.print(CountRun);
  Serial.print(", ");
  Serial.print(StepRun);
  digitalWrite(pinIO[CountRun], LOW);  // On LED
  delay(250);
  digitalWrite(pinIO[CountRun], HIGH); // Off LED
}
```

- XX

22. Test Code “Test0207-Moving LED with 1 Switch”

```
#define SW_L2R 15
#define SW_R2L 4
#define DebounceDelay 100

int pinIO[] = {23, 22, 21, 19, 18, 5, 17, 16};
int CountRun = 0, RunMode = 0;
int minIndex = 0, maxIndex = 7;

void setup() {
  for (int i = 0; i <= 7; i++) {
    pinMode(pinIO[i], OUTPUT);
    digitalWrite(pinIO[i], HIGH); // Active Low
  }
  pinMode(SW_L2R, INPUT_PULLUP);
  pinMode(SW_R2L, INPUT_PULLUP);
  Serial.begin(115200);
}

void SwitchCheck() {
  if (digitalRead(SW_L2R) == LOW) {     // If Switch Press
    delay(DebounceDelay);     // Debounce Delay H->L
    while (digitalRead(SW_L2R) == LOW); // wail until release
    delay(DebounceDelay);     // Debounce Delay L->H
    RunMode = RunMode == 0 ? +1 : 0;
  }

  if (digitalRead(SW_R2L) == LOW) {
    delay(DebounceDelay);
    while (digitalRead(SW_R2L) == LOW);
    delay(DebounceDelay);
    RunMode = RunMode == 0 ? -1 : 0;
  }
}

void loop() {
  CountRun += RunMode;  // Next Step
  CountRun = CountRun < minIndex ? maxIndex : CountRun;
  CountRun = CountRun > maxIndex ? minIndex : CountRun;
  Serial.print("\n > ");
  Serial.print(CountRun);
  Serial.print(", ");
  Serial.print(RunMode);
  digitalWrite(pinIO[CountRun], LOW);  // On LED
  SwitchCheck(); delay(50);
  SwitchCheck(); delay(50);
  SwitchCheck(); delay(50);
  digitalWrite(pinIO[CountRun], HIGH); // Off LED
}
```

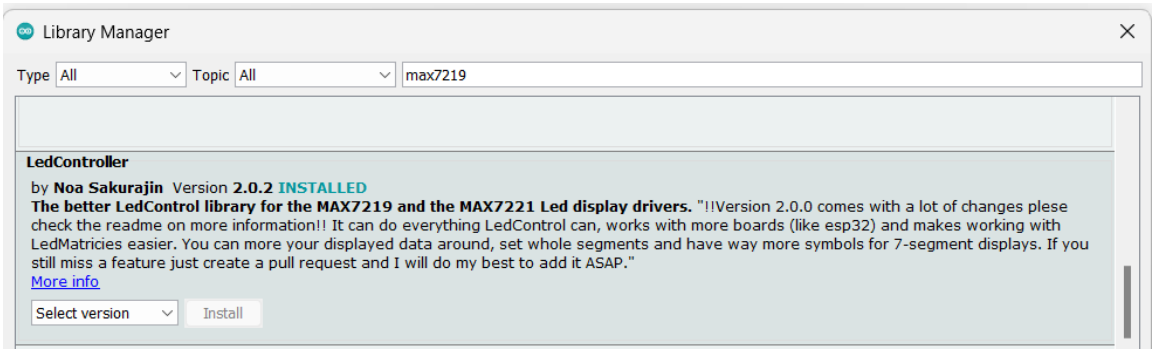
- XX

3/10. ESP32 + Single Digit 7_Segment

4/10. ESP32 + Four Digit 7_Segment

5/10. ESP32 + Eight Digit 7_Segment with MAX_7219

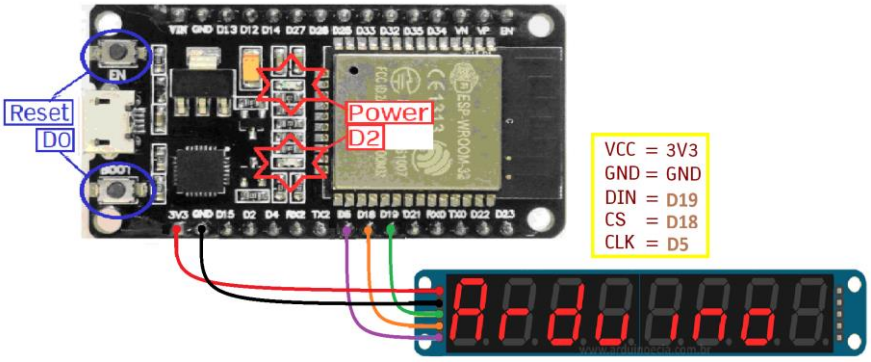
23. Add Library: Sketch → Include Library → Manage
24. Filter with "max7219", Select LedControl by Noa Sakurajin Version 2.0.2



LedController by Noa Sakurajin Version 2.0.2

25. Test Code "Test0501-Hello MAX7219"

```
#include "LedController.hpp"
#define Pin_DIN 19  // D19
#define Pin_CS 18   // D18
#define Pin_CLK 5   // D5

LedController<1, 1> lc;

void setup() {
  lc = LedController<1, 1>(Pin_DIN,
Pin_CLK, Pin_CS);
  lc.setIntensity(8);
  lc.clearMatrix();
}
void loop()
{ lc.setDigit(0, 7, 1, false);
  lc.setDigit(0, 6, 2, false);
  lc.setDigit(0, 5, 3, false);
  lc.setDigit(0, 4, 4, false);
  lc.setDigit(0, 3, 5, false);
  lc.setDigit(0, 2, 6, false);
  lc.setDigit(0, 1, 7, false);
  lc.setDigit(0, 0, 8, false);
  delay(1000);
}
```



26. Test Code "Test0502-Counter"

```
#include "LedController.hpp"
#define Pin_DIN 19  // D19
#define Pin_CS 18   // D18
#define Pin_CLK 5   // D5
long int nCounter = 87654321;
LedController<1, 1> lc;

void setup() {
  lc = LedController<1, 1>(Pin_DIN, Pin_CLK, Pin_CS);
  lc.setIntensity(8);
  lc.clearMatrix();
  Serial.begin(19200);
}
void loop()
{ long int Tempp;
  int singleDigit;
  Tempp = nCounter;
  Serial.println(Tempp);
  singleDigit = Tempp % 10; Tempp /= 10;     lc.setDigit(0, 0, singleDigit, false);
  singleDigit = Tempp % 10; Tempp /= 10;     lc.setDigit(0, 1, singleDigit, false);
  singleDigit = Tempp % 10; Tempp /= 10;     lc.setDigit(0, 2, singleDigit, false);
  singleDigit = Tempp % 10; Tempp /= 10;     lc.setDigit(0, 3, singleDigit, false);
  singleDigit = Tempp % 10; Tempp /= 10;     lc.setDigit(0, 4, singleDigit, false);
  singleDigit = Tempp % 10; Tempp /= 10;     lc.setDigit(0, 5, singleDigit, false);
  singleDigit = Tempp % 10; Tempp /= 10;     lc.setDigit(0, 6, singleDigit, false);
  singleDigit = Tempp % 10; Tempp /= 10;     lc.setDigit(0, 7, singleDigit, false);
  delay(100);
  nCounter--;
}
```

- "lc.setDigit(0, 7, singleDigit, false);"

  0 = Module Numer 0

  7 = Digit 7

  singleDigit = Value Display

  false = no dot display

27. Test Code "Test0503-Switch Counter"

```
#include "LedController.hpp"

#define DebounceDelay   100
#define Counter_Switch  0  // D0 or BOOT
#define Pin_DIN 19  // D19
#define Pin_CS  18  // D18
#define Pin_CLK 5   // D5
int nCounter = 1234;
LedController<1, 1> lc;

void M7219_Display(int Tempp) {
  int singleDigit;
  singleDigit = Tempp % 10; Tempp /= 10;     lc.setDigit(0, 0, singleDigit, false);
  singleDigit = Tempp % 10; Tempp /= 10;     lc.setDigit(0, 1, singleDigit, false);
  singleDigit = Tempp % 10; Tempp /= 10;     lc.setDigit(0, 2, singleDigit, false);
  singleDigit = Tempp % 10; Tempp /= 10;     lc.setDigit(0, 3, singleDigit, false);

  lc.setChar(0,7,'C',false);
  lc.setChar(0,6,'n',false);
  lc.setChar(0,5,'t',false);
  lc.setRow(0,4,B00001001); // tabc defg
}


void setup() {
  pinMode(Counter_Switch, INPUT_PULLUP);
  lc = LedController<1, 1>(Pin_DIN, Pin_CLK, Pin_CS);
  lc.setIntensity(8);
  lc.clearMatrix();
  Serial.begin(115200);
  M7219_Display(nCounter);
}

void loop()
{ if (digitalRead(Counter_Switch) == LOW) {     // If Switch Press
    delay(DebounceDelay);     // Debounce Delay H->L
    while (digitalRead(Counter_Switch) == LOW); // wail until release
    delay(DebounceDelay);     // Debounce Delay L->H
    nCounter++;
    Serial.println(nCounter);
    M7219_Display(nCounter);
  }
}
```

- How are the following commands different?

- lc.setDigit(0, 3, singleDigit, false);

- lc.setChar(0,5,'t',false);

- lc.setRow(0,4,B00001001); // tabc defg

6/10. ESP32 + Dot Matrix with MAX_7219

6.1 ESP32 + Dot Matrix

6.2 ESP32 + Dot Matrix with MAX_7219

7/10. ESP32 + TM_1638

8/10. ESP32 + DHT_22

9/10. ESP32 + Rotary Encoder

10/10. ESP32 + Ambient Sensor